Cloud Infrastructure and Services
Home Assignment - 02

1. Choose an appropiate deployment model for a startup application and justify your choice.

: Chosen Deployment Model: Public Cloud
A startup application should use the public cloud because it is cost-effective, fast to deploy, easy to scale.

Justification:
- No upfront hardware cost (pay only for what you use)
- Easy scaling when users increase
- High availability and reliability
- Built-in security and monitoring
- Faster product launch (important for startups)

E.g. Startups often use platform like Amazon web service or Google Cloud to quickly deploy application without managing servers.

2. Build a basic cloud adoption strategy for an organization migrating from traditional IT environment.

: Basic Cloud Adoption Strategy

1. Assessment
   - Identify existing application, data and infrastructre.
   - Decide which systems are critical and which can move first.

2. Planning
   - Choose cloud provider and cloud type.
   - Define security, compliance, and buget policies.

3. Migration
   - Start with non-critical application (email, backups.)
   - Use lift-and-shift or refactoring based on applica

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4. Optimization
- Monitor performance and cost
- Optimize resource usage.

5. Governance & security
- Apply access controls and encryption
- Regular audits and backups.

3. Select an appropriate cloud service model for hosting a university ERP system.

A university ERP system is best hosted using SaaS.

Reasons:
- No need to manage servers or software updates.
- Easy access for students, faculty, and staff.
- Vendor handles maintenance and security
- Lower operational cost.
- faster deployment.

Example of ERP functions:
- Student information system.
- Fee management
- Attendance and exams
- HR and payroll

Universities often prefers SaaS to focus on education rather than IT maintenance.

4. Apply IDaaS to manage authentication across multiple cloud applications

Identity as a service (IDaaS) provides centralized authentication and access control.

IDaaS is applied to:
- Single Sign-On (SSO) for all application
- Centralized user identity managemen
- Multi-factor Authentication (MFA)

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Benefits:

- One login for multiple cloud apps.
- Improve security
- Easy user management
- Reduced password-related issue

Eg. IDaas Providers:

- Oktas
- Microsoft Azure Active Directory

5. Choose a suitable cloud type for a health care application handling sensitive data.

Healthcare applications deal with highly sensitive patient data, so private cloud is most suitable.

Reason:

- Strong data privacy and control
- Compliance with health care regulations.
- Dedicated infrastructure
- Better access control and security.

Hybrid cloud option:

- Sensitive patient data → private cloud.
- General services (appointments, notifications) → public cloud.

Compliance Example:

Healthcare systems must follow standards like HIPAA to protect patient information.

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